# Python Automata Library (PAL) Cheat Sheet

**Conventions**
import PythonAutomataLibrary as pal · Safe mode is default. Call **pal.FastMode**() before constructing any PAL object. Put substantive model kernels under **@pal.njit(cache=True)**. dims gives grid size per axis: (nx,) for 1D, (nx, ny) for 2D, (nx, ny, nz) for 3D. x... means x, x,y, or x,y,z in 1D/2D/3D. Negative dimensions enable wrapping; out-of-range coordinates wrap on wrapped axes and are skipped on nonwrapped axes.

## Create state

**pal.NewGrid**(dims, dtype): Create a typed lattice storing one value per site.
**pal.NewAgentGrid**(dims, numAgentProps=0, isStackable=False): Create agents with optional properties; use dims=() for nonspatial agents.
**pal.NewPopGrid**(dims, capacity=None): Create integer counts per site; capacity limits total population.
**pal.NewPDEgrid**(dims): Create a continuous field for PDE updates.
**pal.NewIList**(): Create a mutable integer list for collecting/reusing query results.
**pal.NewMultinomial**(): Create a sampler for binomial/multinomial population draws.

## Shared lattice geometry

Properties: **xDim** / **yDim** / **zDim**: axis sizes; **nDims**: number of axes; **wrapX** / **wrapY** / **wrapZ**: wrapping flags.
**ToI**(x...): Convert 1D/2D/3D coordinates into one linear site index.
**ItoX/Y/Z(i)**: Convert a linear site index back to its axis coordinate.
**grid.Hood**(hood, x...): Apply relative offsets at x.... Create hood with **pal.MooreHood**(dim, ...) (diagonals), **pal.VonNeumannHood**(dim, ...) (axis-aligned), or **pal.CircleHood**(dim, rad, ...) (radius). All accept excludeCenter=False.

## Grid / common indexing

Indexing: **grid[i]** reads/writes by linear index; **grid[x...]** uses dimensional coordinates.
Slices: Return detached NumPy copies, not live views; editing a slice does not update the grid.
**Box**(x1,x2,...): Select a rectangular region with lower and exclusive upper bounds for each axis, in axis order.

## Draw / output

**pal.StartPixWindow**(...): Create a pixel buffer and display window; headless=True supports noninteractive rendering.
**pix[x,y] = RGB**: Set a pixel color in the shared pixel buffer.
PixWindow controls: **win.Update**() refreshes the display; **win.Save**(path,block=True) saves an image; **win.Close**() releases the window.
PixWindow GIF: **win.StartGif**(path,delay=100) starts recording; **win.AddGifFrame**(block=False) appends a frame; **win.StopGif**() finishes.
**pal.StartOpenGLWindow**(xDim,yDim,zDim=None,width=800,height=800,...): Create the renderer and return (draw, win): draw collects primitives; win handles display, updates, and output. Omit zDim for 2D; set it for 3D. headless=True supports noninteractive rendering.
**draw.Circle**(rad,color,x,y,z=None): Add a circle or sphere-like primitive with the given radius and color.
**draw.Box**(xLen,color,x,y,z=None,yLen=None,zLen=None): Add a box centered at a position; specify axis lengths to control its size.
**draw.BoxSQ**(color,x,y,z=None): Add a box aligned to a lattice site.
**draw.Line**(width,color,x1,y1,x2,y2,z1=None,z2=None): Add a line segment between two 2D/3D points.
**draw.Borders**(width,color): Draw grid borders with the requested line width and color.
**win.Camera**(x,y,z,yaw=None,pitch=None): Set the camera position and optional yaw/pitch angles for a 3D scene.
OpenGL frame: **win.Background**(color) sets the background; **draw.Clear**() clears collected primitives; **win.Clear**() clears the scene; **win.Update**() renders.
OpenGL output: **win.IsOpen**() checks the window; **win.Save**(path,block=False) saves an image; **win.Close**() releases resources.
OpenGL GIF: **win.StartGif**(...) begins recording; **win.AddGifFrame**(...) adds a frame; **win.StopGif**(...) finishes.

## Lists + randomness

IList methods: **Append**(i) adds an integer; **Clear**() empties the list; **Random**() picks an entry; **Shuffle**() reorders entries; **All**() copies the list; **Iter**() traverses without copying.
RNG methods: **pal.Seed**(seed) seeds PAL's RNG; **pal.Random**() draws uniform [0,1); **pal.RandInt**(n) draws an integer in [0,n). PAL's RNG is separate from NumPy's.
Multinomial methods: **m.Binomial**(n,p) draws a binomial count; **m.Setup**(...) configures multinomial probabilities; **m.Sample**(...) draws counts.

## AgentGrid

**NewAgentSQ**(x...) / **MoveSQ**(a,x...): Create or move an agent using lattice-site coordinates; documented linear-site forms are also supported.
**NewAgent**(x...) / **Move**(a,x...): Create or move an agent using continuous coordinates rather than lattice sites.
**grid[a,p]**: Read or write property p for agent a.
Agent position: **I**(a) gives the linear site index; XSQ/YSQ/ZSQ(a) give lattice coordinates; X/Y/Z(a) give continuous coordinates.
**Alive**(a) / **Dispose**(a): Check whether an agent is still valid, or remove it from the grid.
**GetPop**(): Count agents currently in the grid.
**AgentsAt**(x...) / **LastAgent**(x...): Find agents at a site or retrieve the last agent there.
**AgentsInRadius**(...): Find nearby agents, respecting periodic boundaries.
**counts[x...]**: Read site occupancy counts without modifying them.
Iteration: **agents.All**() returns a detached snapshot, safe for iteration while adding or disposing agents.
**DispWrapX/Y/Z(p1,p2)**: Compute the shortest displacement along a periodic axis.

## PopGrid + PDEgrid

Synchronous updates: **Add**(v,...) queues a change; **Update**() applies queued changes together; **Reset**() clears current and pending values. Indexed assignment changes current state immediately.
**pop.GetPop**(): Return total population across all sites.
**pop.All**(): Return indices of sites with nonzero population, not the population values.
PDE setup: **field.SetTimeSpaceStep**(dt, dx[,dy,dz]) sets time and spatial steps before updates. Use spacing values that satisfy the selected scheme's stability requirements.
Diffusion methods: **Diffusion**(...) uses the standard scheme; **DiffusionMask**(...) applies a mask; **DiffusionField**(...) uses spatially varying coefficients; **DiffusionInterfaces**(...) specifies interfaces; **DiffusionADI**(...) uses an implicit scheme; DiffusionRadialCircle/Sphere(...) handles radial geometry.
Advection methods: **Advection**(...) uses constant velocity; **AdvectionField**(...) uses spatially varying velocity; **AdvectionInterfaces**(...) specifies interface velocities.

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**More detail: Manual · API Guide · API Reference (Documentation/).**